

Invited talk · Measuring and modeling transcription factor dosage effects

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Transcription factors (TFs) dictate cellular and developmental gene expression programs by binding to specific DNA motifs within noncoding regulatory elements (REs) and modulating their activity. Quantitative changes in TF levels (dosage) underlie both common phenotypic variation and severe disease, but the mechanisms underlying this TF dosage sensitivity have remained poorly understood. I will describe our efforts to tackle this problem using experimental and computational approaches. First, we combined chemical genetic tools, stem cell models of human neural crest cells, and epigenomic mapping to quantify how REs and genes respond to precisely modulated TF dosage. We found that while many TF-dependent REs and genes are buffered against small changes in TF dosage, others are highly sensitive, linking altered TF dosage to specific developmental phenotypes. Second, we applied sequence-to-activity deep learning models to predict RE responses from DNA sequence, achieving near-experimental accuracy. These models revealed predictive sequence features that are undetectable by models of steady-state RE activity, and could interpret noncoding variants associated with variation in craniofacial shape. These results show that combining quantitative TF perturbations with deep learning can uncover additional layers of the cis-regulatory code, with implications for the development of cellular foundation models.